



Asteroid family classification with machine learning: Investigative analysis of a novel two-step approach for categorizing known small asteroid families*

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Abstract

The term “asteroid family” refers to a collection of asteroids that share similar proper orbital elements such as semi-major axis, eccentricities, and orbital inclinations. Detecting small asteroid families has proved to be a challenge for a long time because of their extremely low sample size. In general, standalone machine learning classifiers tend to exhibit a bias towards classes with larger sample sizes, resulting in the inadequate classification of asteroid families with limited data. In this paper, a two-step supervised model was proposed for the effective classification of the asteroid families, especially for the tiny, small, and lower groups of medium asteroid families. The proposed model uses two-step classification in an attempt to resolve the challenges that come with the imbalanced dataset where at first a binary classification of small and large families was done with an XGB (Extreme Gradient boosting) classifier and then in the second stage Random Forest classifier was used alongside previously identified binary features to classify asteroid families. The proposed model performed better with higher F1 scores for tiny and small asteroid families compared to other algorithms tested in this work. It also achieved a perfect F1 score for 90 families, among 112 families which were tested. As for the lower group of medium sized asteroid families, it performed slightly worse compared to the previously used machine learning algorithms. Along with this, four dataset imbalance handling techniques have been employed in this work and compared to the proposed algorithm.

Keywords Asteroid family · Machine learning · Classification · Two-step model · XGBoost · Imbalanced learning · F1-score

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1 Introduction

Asteroids are the remainder of the solar system. During the formation period, the presence of Jupiter prevented the emergence of a new rocky planetary body between the gap between Mars and Jupiter. The residue of the unformed planet and the other deposited bodies remained stasis in the area, often colliding with one another, now known as asteroids. Asteroids occur in a variety of shapes and sizes, which provides insight into their genesis in the early solar system. Asteroids are essentially monuments in the history of the solar system. Therefore, the study of asteroids has been the center of attraction for many contemporary researchers. In the context of asteroids, it is important to study asteroid families, which are collections of asteroids that have a similar origin and share similar proper orbital elements such as semi-major axis, eccentricities, orbital inclinations, etc. The accurate classification of asteroid families is important to know their detailed properties, hence getting a better insight into the formation period of the solar system. There have been a handful of methods and notable works regarding asteroid family classifications.

The foundational work of [1] was the first to systematically identify asteroid families, which he referred to as ‘groups of asteroids probably of common origin.’ By analyzing proper orbital elements such as semi-major axis, eccentricity, and inclination, Hirayama established that certain asteroids share a likely common origin, possibly due to the fragmentation of a larger parent body. This pioneering study laid the groundwork for subsequent advancements in asteroid classification techniques and remains a cornerstone in planetary science research. Subdivision of Flora families through a subjective criterion [2] and Poisson Distribution [3] are some of the notable works. Works with esoteric methods (complex mathematical and statistical analysis) such as the Hierarchical Clustering Method (HCM) [4], which has been the most widely used and preferred method for asteroid family classification since the 1990s because of its ability to group asteroids based on their proximity in space of proper orbital elements, offering a straightforward yet powerful way to detect families within large populations. In addition to HCM, other methods such as the Wavelet Analysis Method (WAM) [5], D criterion [6], have historically performed well for categorizing asteroid families.

The works of [7] were among the first to adopt these techniques for asteroid family classification, with subsequent refinements and widespread application seen in their later work [4], where the parameters for defining a family were: proper semi-major axis (a), eccentricity (e), and inclination (i). Another notable work that utilizes an artificial neural network (ANN) was proposed by [8] which used the AstDys dataset to train their model. In the works of [9] the possibility of using 9 supervised machine learning models for identifying new asteroid family members was explored. It is clear from the results of this article that the classification of medium, small, and tiny asteroid families is the most difficult as there are very few samples to work with. That is why in this paper a model has been proposed that works in two steps to identify tiny, small, and lower groups of medium asteroid families. Asteroid families are classified based on their member count: tiny families have fewer than 30 members, small families have between 30 and 100 members, and medium families have between 100 and 1000 members [10].

The examination of prior scholarly works reveals that machine learning methods exhibit a notable deficiency in effectively classifying smaller asteroid groups. One primary factor contributing to this issue is the presence of class imbalance, which leads to machine learning models exhibiting a bias towards classes with greater sample sizes. Traditional machine learning algorithms can recognize the patterns of the data precisely when they are trained on an almost equal number of positive and negative class examples. Whenever there is a class imbalance in the dataset, the algorithms often get biased towards the majority class and fail to identify the minority class examples correctly. Class imbalance is found in most real-world datasets and machine learning models often show a bias. Hence, handling imbalanced data has become an important issue for solving real-world problems. Researchers have developed various approaches to handle the class imbalance problem [11]. Balancing the dataset is one of the widely used techniques to address class imbalance. This method is a part of data-level preprocessing. It is also known as data sampling. In this technique, the training data is resampled in a way that an almost equal number of instances are produced in each training class.

Classification algorithms trained in these resampled data perform much better than algorithms trained on an imbalanced dataset. Sampling techniques can be further divided into two main categories oversampling and undersampling. In oversampling techniques, minority class examples are resampled to produce new instances from existing examples. Smote is one of the most popular oversampling techniques [12]. Unlike Random oversampling, it produces synthetic samples from the existing ones. In undersampling techniques, a few of the redundant majority class examples are eliminated to balance the dataset. Oversampling and undersampling methods can be combined to produce hybrid sampling techniques and improved performance can also be obtained using hybrid sampling [11]. Cost-sensitive learning is another technique to handle imbalanced datasets [13]. It is a part of algorithmic-level modifications. In traditional classification algorithms, misclassification cost is considered the same for both majority and minority class examples. To reduce the overall cost, the algorithm tries to predict the negative class examples perfectly as those are majority class examples. As a result, the model fails to show a better performance for the minority class. In cost-sensitive learning, different misclassification costs are assigned to majority and minority classes. For minority class examples, a higher misclassification cost is assigned so that the bias is shifted towards the positive class. The assigned cost depends on the ratio between the number of majority and minority examples commonly known as the Imbalanced Ratio (IR). Usually, a higher cost is assigned for a higher Imbalance Ratio (IR). Ensemble learning algorithms are also used to address class imbalance [14]. Ensemble learning algorithms are the combinations of two or more machine learning algorithms. Balanced Random Forest is the most renowned ensemble learning algorithm to handle imbalanced datasets [15].

In this work, a new dataset has been created using the AFP database [16]. Supervised machine-learning algorithms have been utilized to classify asteroid families. The supervised models employed in this paper use proper semi-major axis (a), proper eccentricity (e), and sine of proper inclination ($\sin(i)$) to classify the families of the asteroids. The proper semi-major axis (a) represents the size of the orbit of an asteroid. Asteroids with the same origin tend to have a close value of proper semi-major axis. Proper eccentricity (e) represents the shape of the orbit and like semi-major axis asteroids with similar origins also tend to have close eccentricity. Finally, the inclination (i) represents the angle

of tilt of the orbit of an asteroid concerning the ecliptic plane. While collisions can alter orbital elements such as inclination, they generally do not cause drastic changes. This relative stability allows for the identification of asteroid families based on the clustering of their proper orbital elements following collision events.

Furthermore, this study introduces a binary classification architecture based on a ‘member index,’ where asteroid families with more than 250 members are assigned a value of 1 (significant sample size), and those with fewer members are assigned a value of 0. The member index is subsequently used in the multi-class classification phase to improve accuracy and handle class imbalance. In the first step instead of assigning modified weights for misclassification, the proposed model adds weights as features in the dataset. Here a new feature has been added named ‘Member-Index’ which works as a weight for each instance. Because of the assigned weight, the traditional classification algorithms show better performance when trained by the modified data. The proposed model performance is compared with the performance of other imbalance data handling techniques such as Smote, Random Undersampling (RUS), cost-sensitive Random Forest, and Balanced Random Forest (BRF) in Sect. 5.5. The comparison shows the proposed model performed better than all other techniques used to address imbalanced data. Among the standalone models, the best algorithm turned out to be the Gaussian Naïve Bayes algorithm in terms of the best F1 score, with a weighted average score of 0.95172. The proposed architecture did somewhat better than that with a weighted average F1 score of 0.95184.

The paper is organized as follows: In Sect. 2, all the related works and ongoing research have been reviewed. An overview of the dataset, related visualization, and the proposed model have been represented in Sects. 3 and 4. In Sect. 5, the result and related error analysis have been discussed. Finally, Sect. 6 concludes the paper along with possible future works.

2 Related works

The evolution of asteroid family classification witnessed a paradigm shift with the integration of diverse machine-learning methodologies. Zappalà et al. [4] pioneered this domain in the early 1990s, introducing the hierarchical clustering method (HCM). This method employed a tree-like architecture, merging the closest objects of a given sample until only one entity remained. Utilizing parameters such as the semi-major axis (a), eccentricity (e), and inclination [4] established a three-dimensional space, successfully distinguishing 26 distinct families, each consisting of a minimum of 5 members.

In parallel, Bendjoya et al. [5] presented the wavelet cluster analysis (WAM, a specialized technique for density evaluation based on specific wavelet functions). This method facilitated the identification of proximate points in an N -dimensional space, where each point represented an asteroid characterized by its properties: a , e , and i . The WAM methodology achieved point grouping. As the wavelet coefficient increased, classification precision enhanced. Although both HCM and WAM surpassed preceding methodologies in efficacy, the simplicity of HCM made it favorable.

Gehrels et al. [6] introduced the D criterion, a distinct approach utilizing computational analysis. This technique revolved around a neighbor-linking algorithm that clustered points based on predetermined metrics and a specific cut-off range, ensuring the weeding out of inaccuracies. Contrasting the HCM and WAM, the D criterion incorporated a unique attribute definition as a metric. Initial applications of these methodologies analyzed a dataset comprising approximately 4000 proper elements sourced from [10] proper-element dataset. To gain stability and performance relative to parameter variations, Zappalà et al. [4] introduced robustness coefficients for each grouping. The classifications were often complicated by intersecting families, particularly prominent in the inner belt, exemplified by the Polana family [5]; [6]; [4].

Studying the spectral domain, Nesvorný et al. [17] investigated color variations among asteroid families. Their research evidenced processes that instigate color alterations in asteroids over chronological scales. Leveraging the Sloan Digital Sky Survey (SDSS) catalog, they discerned that as S-type asteroids age, their surfaces progressively adopt redder hues. Further refining the SDSS methodology, [18] incorporated albedos as the fifth dimension, considerably diminishing known interlopers. This enhancement was later corroborated by Novaković et al. [16].

Novaković et al. [19] expanded on high-inclination asteroid families, exploring classification challenges associated with such groups. Their study pointed out the struggles that traditional clustering methods have in recognizing these families and suggested better approaches to tackle the problem.

Milani et al. [20] unveiled a novel classification technique, mixing the Hierarchical Clustering Method (HCM) with an innovative method to incorporate new members into pre-existing asteroid families. By implementing a threshold for Absolute magnitude (H limit), core family entities were identified. If an asteroid's proximity to a core family member didn't exceed the critical distance, it was attributed to the core family. If an asteroid aligned with multiple families, it was categorized under intersections.

Subsequently, Milani et al. [21] expanded their analysis to include multi-opposition objects, culminating in the classification of over 500,000 asteroids. This extension adeptly addressed dual challenges: pinpointing new families and attributing new asteroids to established families. Diving into the Yarkovsky effect, as elaborated by Walsh et al. [22], asteroid families, when graphically represented against their semi-major axis and inverse diameter (or absolute magnitude), exhibited a distinctive V-shaped pattern. This pattern's breadth is related to the rate at which the asteroid's semi-major axis shifts due to the Yarkovsky effect.

Artificial neural networks (ANN), furnish several advantages for cluster detection tasks. A testament to ANN's efficacy is done by Vujičić et al. [8] who employed the AstDys database consisting of 124,478 members and 110 cluster families. Their methodology comprised 56 hidden layer neurons, coupled with a ten-fold cross-validation. The resultant accuracy for training data ranged between 98.82% and 99.33%. Here, the standard deviation was 0.17%. However, small families were having trouble in this case. 0% testing accuracy was found in six of 46 families where the number of members (in those families) was less than 100. Families having 101 to 1000 members gave satisfactory results. As a result, it is clear that effective classification of small and tiny families with fewer than 100 members remains a challenging task.

Vokrouhlický et al. [23] proposed an innovative “box” method for identifying members of young asteroid families. They took Yarkovsky effect and resonant interactions into account. This technique uses both orbital elements and dynamical constraints to accurately classify younger families. This technique especially excels for those that have undergone fewer collisional and dynamical interactions. Their approach offered a novel way to segregate young families from background asteroids. [16]

Carruba’s latest study [9] adeptly leveraged ML to pinpoint new asteroid families. Utilizing bootstrap samples for data processing, the study discerned that every decision tree technique inherently bore some redundancy. The outcomes were promising, with Extra Tree outperforming other methods for families where $H < 14$. Notably, the KNN approach excelled for small to mid-sized families. Subsequent validation led to the 0–1 classification of four specific families. Following evidence that it was a more effective strategy, 4 families – (694) Ekard (105 members), (480) Hansa (1484 members), (15) Eunomia (6076 members), and (832) Karin (480 members) – were given the 0–1 classification. This section furnishes a comprehensive review of prior works, instrumental in shaping the ground for asteroid family classification, and clarifies the reason behind choosing the small asteroid family classification.

3 Dataset

The dataset used for this study was the Asteroid Families Portal dataset, maintained by the Department of Astronomy, University of Belgrade (<http://asteroids.matf.bg.ac.rs/fam/properelements.php>) as it has one of the most updated collections of 793,310 asteroids classified in 112 families (as of January 2021). The AFP dataset contains catalogs of asteroids in both the main belt and the trans-Neptunian region which was cataloged and categorized using the synthetic theory [24].

This AFP dataset contains various physical and dynamical properties of asteroids. Some datasets have spectroscopy color indexes, such as the Sloan Digital Sky Survey [25] and albedo, such as WISE [26]. However, these datasets have a smaller number of asteroids cataloged, including observations with poor accuracy. As stated in the works of A. Milani [20] about the AFP dataset, it is a rapidly growing dataset with about a 5.4% increase every 5 months. Since this study aims to utilize machine learning algorithms for asteroid classification, the large catalog of this dataset proved beneficial for training and testing. Also, the growing number of cataloged asteroids can be used to further train the models for improved reliability of classification in the future.

3.1 Overview of the dataset

The AFP dataset contains 10 columns that depict 9 features along with the asteroid designation number. The columns are as follows:

Column 1: Asteroid designation number.

Column 2: Absolute magnitude (H).

Column 3: Proper semi-major axis (a).
Column 4: Proper eccentricity (e).
Column 5: Sine of proper inclination ($\sin(i)$).
Column 6: Proper frequency of mean motion [deg/yr].
Column 7: Proper frequency of perihelion longitude [arcsec/yr].
Column 8: Proper frequency of nodal longitude [arcsec/yr].
Column 9: Lyapunov time (LCE [kyr]).
Column 10: Integration time span [Myr].

In this study, the proper semi-major axis (a), proper eccentricity (e), and $\sin$ of proper inclination ($\sin(i)$) were used as the features for training. These three are the dynamic properties or proper elements of an asteroid that have a much larger information content [10].

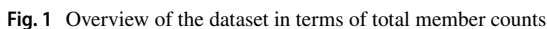
The proper semi-major axis (a) represents the size of the orbit of an asteroid. Asteroids with the same origin tend to have a close value of proper semi-major axis. proper eccentricity (e), represents the shape of the orbit, and like semi-major axis asteroids with similar origin also tend to have close eccentricity. Finally, the inclination (i) represents the angle of tilt of the orbit of an asteroid with respect to the ecliptic plane. The inclination is also a descriptor of past collisions as collisions can change the inclination. In this dataset, the sine of inclination was used because it provides a more symmetrical distribution of the inclination values which can facilitate the training of the models in this study. Contrary to the works of [8], absolute magnitude (H) was not chosen in this study because absolute magnitude for asteroids is calculated by averaging inhomogeneous observation data that often is poorly controlled for errors [20].

Along with the existing columns, two more columns were added, the first one is Dataset name which was the family name of the asteroids, that was used as the label. The member index is a binary variable used to classify asteroid families based on their size. Families with 250 or more asteroids are assigned a value of 1, indicating a significant sample size, while families with fewer than 250 asteroids are assigned a value of 0. The Member index feature assigns weightage to prevent misclassification. The cut-off point is necessary. We have examined the dataset, and assigned the cut-off point accordingly to keep the noise minimum. For further sub sequent multi-class classification, the cut-off point can be changed accordingly.

3.2 Visualization of the dataset

Figure 1: The plot illustrates the distribution of asteroid family sizes based on member counts. The top panel highlights families with member counts exceeding the classification threshold of 250 (yellow box), which are assigned a binary value of 1, while smaller families are assigned a value of 0. The bottom panel provides an overall view, emphasizing the steep drop-off in smaller family sizes, which highlights the challenges of classifying families with sparse memberships and the importance of addressing size diversity in the classification process.

Figure 2 shows proper semi-major axis vs. sine of proper inclination which visualizes a clear grouping between various asteroids, 35 groups in total. This



distribution is similar to the ones in the works of D. Vujičić, in which the features of proper semi-major axis and sine of proper inclination were also used. The groups tend to reside within 2.5 to 3.5 AU on the scale of proper semi-major axis, with only 3 groups outside this range.

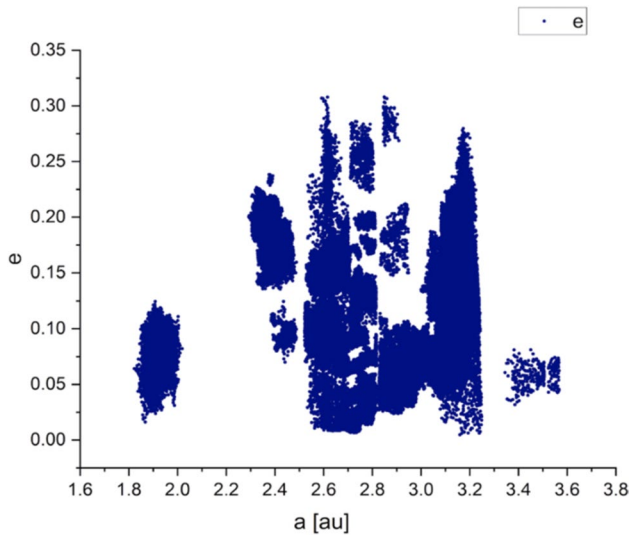


Fig. 3 Proper semi-major Axis vs. proper eccentricity

Similarly, Fig. 3 shows the proper semi-major Axis vs. proper eccentricity. In this graph, the majority of the grouping lies between 2.4 and 3.2 AU on the scale of the Proper semi-major axis, which indicates that this Semi-Major Axis range is a good subset for training the models. The works of D. Vujičić had a similar approach to identifying a training subset, where they used Proper semi-major Axis vs. Absolute Magnitude distribution.

3.3 Limitations of the dataset

The AFP dataset used in this work contains some easy-to-measure features of asteroids that describe the physical and dynamic properties of the asteroids. However, the features from the AFP dataset that were used in this work have their limitations. This is not an issue of the dataset, but rather the natural observations of the physical and dynamical properties of asteroids show variability. sine of proper inclination ($\sin(i)$) can show variability, ranging from 0 to 0.4 within the same families as well as having nearly the same values for different families (low-incline main belt families) which is natural to occur in the observation of asteroids. proper eccentricity (e) and Proper semi-major Axis (a) can also have close values for asteroids that have overlapping orbits. This variability, mentioned as instability of proper elements in the works of [20] shows that it can potentially throw off classification algorithms for asteroids with very similar properties.

Limitations like these are to persist in any dataset that catalogs the physical and dynamic properties of asteroids. However, despite the potential limitations, the AFP dataset has a large catalog, is regularly updated, and is supported by the well-established work of A. Milani [20], and therefore, was chosen for this study.

4 Methodology

This section has covered the various machine-learning classification strategies used to categorize asteroid families. For this, the suggested model along with seven machine learning methods was employed. The algorithms used were Gaussian Naïve Bayes, Logistic Regression, Extra Tree Classifier, Random Forest Classifier, Decision Tree Classifier, AdaBoost Classifier, and KNN Classifier. XGBoost classifier was used for the binary classification.

4.1 Two-step supervised model

The proposed algorithm tackles the problem associated with classifying smaller families. A singular classifier often biases toward families with larger numbers of members which can be seen in Tables 5 and 6. Various dataset imbalance handling techniques have been investigated in this study such as RUS, oversampling, using weights for cost-sensitive learning, etc. However, the case where weights are used as features are not well utilized in previous work and this study aims to implement this technique. Therefore, it was required that another parameter be introduced to mitigate imbalance bias. In the works of [20] it could be seen that smaller asteroid families were misclassified the most which inspired us to put weights on these families and hence segregate large and small families. When any new data is received then information regarding proper semi-major axis (a), proper eccentricity (e), and sine of proper inclination ($\sin(i)$) are only present and so if this extra feature is to be added to the dataset to represent the weights based on family size, it is needed to predict if the asteroid belongs to an asteroid family with small sample size. Therefore, a binary classification label was introduced first in the training data set. In that binary classification label families with more than 250 members were identified with the value of 1, and families with less than 250 members were identified with the value of 0. In the literary works of [20] tiny asteroid families are those that have fewer than 30 members, small asteroid families are those that have more than 30 members but less than 100, and medium asteroid families are those that have more than 100 members but fewer than 1000. Since this research focuses on asteroid families that are either tiny, small or fall into the lower range of medium asteroid families, the number 250 was used for convenience. In this approach, the XGB classifier was trained with proper semi-major axis (a), proper eccentricity (e), and sine of proper inclination ($\sin(i)$) to classify the asteroid families if the sample size was over 250 or less than 250. The proper semi-major axis (a), proper eccentricity (e), and sine of proper inclination ($\sin(i)$) in the training dataset along with the binary classification label or `member_index` feature was then used to train the Random Forest classifier to classify the asteroid families. The testing data was fed into the XGB model after being trained and as a result of that a new feature was added to the testing data which indicated the weight assigned to a particular asteroid. This updated testing set was then fed into the Random Forest classifier which was trained to classify the asteroid families. The following flowchart, Fig. 4, illustrates the preparation, train, and test process of the proposed model.

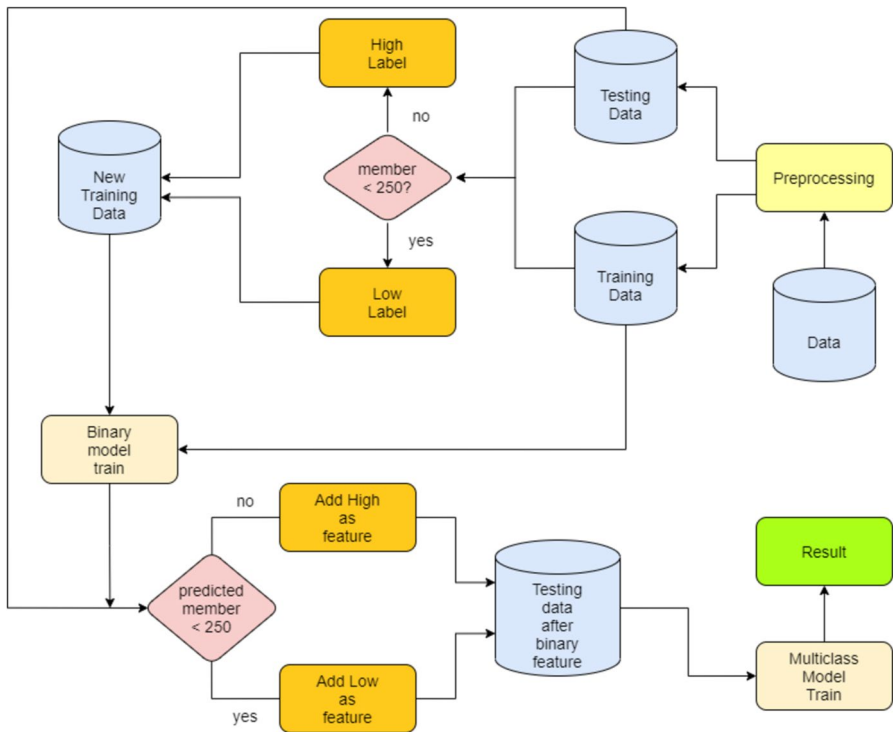


Fig. 4 Flow chart of proposed algorithm

It is to be noted that these two algorithms were chosen for the two-step supervised model as this combination performed the best among all other combinations which were tested in the trial-and-error phase.

4.2 Standalone models

To compare the results of the proposed model with other known machine learning architectures we also trained decision tree, random forest, extra tree, logistic regression, gaussian naïve bayes, KNN, and adaptive boosting models to classify the asteroid families. These standalone models were chosen since these models were used previously in the works of [9]. Choosing such well-known models made it easier to compare the proposed two-step supervised model with a stable baseline.

4.2.1 Decision tree

A decision tree is a classification algorithm structured in a tree-like manner to determine a target result. The training data initiates at the root node. At each decision node, where branching occurs, statistical methods are employed to optimally split attributes

among the children, and an initial classifier value is deduced. The branching algorithm employed is the ID3 (Iterative Dichotomiser 3), which follows a greedy strategy without backtracking [27]. Attributes are chosen based on multiple criteria: entropy, information gain, Gini ratio, Gini index, chi-square, and reduction in variance.

4.2.2 Random forest

Random forest adopts the same principle as decision trees but incorporates multiple trees in an ensemble, with each tree operating independently of the others' error rates. The consensus outcome from the majority of the trees determines the final result of the algorithm. The ensemble workflow of random forest typically ensures enhanced accuracy [28].

4.2.3 Extra tree

The Extremely Randomized Tree (EXTRA Tree) algorithm, akin to the Random Forest, operates as an ensemble classifier [29]. Each tree concurrently selects the best feature set from its specific subset using criteria like the Gini Index. Due to randomized distribution, the decision trees are decoupled in processing enigmas. The overall algorithm pinpoints a particular feature based on the maximized criterion of the decision trees.

4.2.4 Logistic regression

Logistic regression is a categorical data prediction algorithm that employs the sigmoid function over the yielded prediction value to classify the prediction between two categories. If the sigmoid value approaches infinity, the predicted category is 1, and if it heads towards negative infinity, the category is 0. The mean squared error function serves as the cost function. A lower cost function value indicates a more precise fit [30].

4.2.5 Gaussian naïve bayes

The Gaussian naïve Bayes algorithm assumes feature variables to be independent and postulates that they follow a Gaussian distribution. For each data point, the z-score is computed about the mean of each feature set, determining the likelihood of the point belonging to a specific class [31].

4.2.6 K nearest neighbors

The K nearest neighbors (KNN) algorithm gauges the proximity of data points to each other, considering a predefined threshold of K neighbors. Data points within this proximity threshold are collectively classified as a singular group [32].

4.2.7 Adaptive boosting (AdaBoost)

AdaBoost is an ensemble classification technique employing numerous decision trees to adaptively process datasets. Initially, each data point is assigned equal weight. Subsequently, decision stumps (single-split decision trees) are created. The number of these initial stamps corresponds to the number of features on these data points. A base learner is selected based on the lowest Gini and Entropy. Misclassified data points receive higher weights, adjusted based on their performance and the overall error index. This iterative process continues until the error reaches an acceptable minimum. Finally, the classifications from all iterations are combined to form the final output [33].

4.2.8 Extreme gradient boosting (XGB)

Extreme Gradient Boosting or XGB utilizes a gradient boosting framework over a decision tree-based approach for classification [34]. The XGB classifier sequentially selects features and adopts a depth-first tree pruning method with a max-depth parameter, optimizing computational efficiency. Additionally, XGB incorporates automatic cross-validation and sparsity awareness, refining classification accuracy.

4.3 Data preprocessing

Using the information from the AFP website, a dataset with the features a , e , and $\sin(i)$ was created. The data was then split into training and testing groups. 80% of the data was used for training, and the remaining 20% was used for testing. For creating the training and testing dataset stratified method was used to eliminate sampling bias in the dataset. Train-test-split function follows a random selection method to split the dataset into training and test set. This function does not consider the label of the data when splitting. As a result, the imbalance ratio does not remain same in the training and test set. Stratified train-test-split divides the dataset in such a way that it preserves the same ratio of positive and negative samples in the training and test set.

Here all the hyper-parameter optimization was done using only the training dataset. The testing dataset was only used for the final evaluation. A binary member index column was added to the dataset which indicated if an asteroid had more or less than 250 samples for its asteroid family. It was discovered that 56 asteroid families were such which had less than 250 members in its family. The minority oversampling technique smote was used on the dataset to perform the binary classification. Regarding the multi-class classification, no oversampling technique was used.

For tiny families ($n < 30$), the size of the test dataset (fewer than six members) may lead to variability in metrics such as precision, recall, and F1-score. This is due to the limited representation of these families in the evaluation phase, making the results more sensitive to individual misclassifications. While these metrics provide valuable insights, they should be interpreted cautiously, as small test datasets may not offer statistically robust performance evaluations.

5 Results

In this part, all metrics utilized to evaluate the results will be discussed. In addition, the produced result will be compared to earlier obtained results.

5.1 Hyperparameter tuning of the tested models and proposed two-step model

In the experiment, all the tested models and the two-step models were hyperparameter-tuned with Randomized Search cross-validation (CV). The Random Search CV is a parameter-tuning algorithm that evaluates the results obtained from a fixed number of combinations of the parameters that were passed to the algorithm. The reason Randomized Search CV was used was because the objective was to determine a parameter value range rather than selecting a singular value. The training set was only used for optimizing the hyper-parameters. This algorithm first splits this dataset into training set and validation the size of which depends on the cross-validation value which is set prior to running the algorithm. In this study, 3-fold cross-validation has been used for optimizing the hyper-parameters. The parameter values have very little value difference on an individual scale; hence value range was selected for parameter tuning. Every tested parameter value is representative of a value range. The result of hyperparameter tuning is shown in Table 1.

5.2 Evaluation metrics

5.2.1 Precision

Precision is the percentage/fraction of correctly predicted true positive values in contrast to all predicted positive values. [35] This is calculated by dividing the number of correctly predicted positive values by the number of all predicted positive values (number of predicted true positive values + number of predicted false positive values). For example, in the algorithm, precision measures the number of total true positives for an asteroid family in contrast to the total positives of that family. The equation of precision is as follows:

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

where TP is the number of true positives and FP is the number of false positives. Table 2 visualizes the relation between true positive (TP) and false positive (FP).

Here, TP represents True Positive and TN represents True Negative.

5.2.2 Recall

Recall is the percentage/fraction of the number of correctly predicted positive values against the total number of predicted positive values. [36] : Recall measures the proportion of correctly identified positive instances out of all actual positive instances, effectively capturing the model's ability to identify relevant data. This is calculated

Table 1 Hyperparameter tuning of the tested models and proposed two-step model

Model	Hyperparameter	Ranges	Best Value
Decision Tree	minimum sample split	2, 4, 6, 10	2
	max_depth	3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 16, 25, 50, 100, 200	12
Random Forest	number of estimators	40, 60, 65, 70, 75, 80, 85, 90, 95, 100, 120	40
Extremely Randomized Tree (EXTRA Tree)	minimum sample split	2, 4, 8, 10, 15, 20, 30	20
	number of estimators	50, 126, 25	50
	minimum sample split	15, 25, 35[c]	25
	minimum sample leaf	5, 20, 50	20
	Optimization algorithm, solver	'lbfgs', 'liblinear', 'newton-cg', 'newton-cg-cholesky', 'sag', 'saga'	newton-cg
Logistic Regression	regularization parameter, C	0.1, 1, 10	10
	Tolerance for stopping criteria, tol	0.1, 0.01, 0.001, 0.0001	0.1
Gaussian Naïve Bayes	var_smoothing	1,2,3,4,5,6	#
K nearest neighbors (KNN)	number of neighbors, n_neighbors	1,2,3,4,5	2
	leaf size	15,30,50,100	100
Adaptive Boosting (AdaBoost)	Power parameter, p	1,2,3	3
	number of estimators	2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 20	3
	learning rate	0.97, 0.98, 0.99, 1, 1.01, 1.02, 1.03, 1.04	0.9
	Optimization algorithm	SAMME, SAMME.R	SAMME

Table 1 (continued)

Model	Hyperparameter	Ranges	Best Value
Extreme Gradient Boosting (XGB)	L1 regularization term on weights, α	0.1, 0.5, 1, 2, 3, 4, 5	1
	Step size shrinkage, η	0.001, 0.1, 0.3, 0.5, 1, 3, 5[e]	0.1
	L2 regularization term on weights, λ	0.1, 0.5, 1, 3, 5,	0.5
	subsample ratio of columns for each tree, colsample-bytree	0.1, 0.5, 0.8, 1	1
	Subsample ratio of the training instances, subsample	0.1, 0.5, 1	1

Table 2 Relation between the elements of the confusion matrix

Actual Class	Predicted Class	
	Positive	Negative
Positive	<i>TP</i>	<i>FN</i>
Negative	<i>FP</i>	<i>TN</i>
Metric	Formula	
<i>Precision</i>	$\frac{TP}{TP+FP}$	
<i>Negative Predictive Value</i>	$\frac{TN}{TN+FN}$	
<i>Accuracy</i>	$\frac{TP+TN}{TP+TN+FP+FN}$	
<i>Secificity</i>	$\frac{TN}{TN+FP}$	
<i>Sensitivity (Recall)</i>	$\frac{TP}{TP+FN}$	
<i>False Positive Rate</i>	$\frac{FP}{FP+TN}$	

Here, *TP* represents True Positive and *TN* represents True Negative

by dividing the total number of correctly predicted positives by the number of all predicted positive values (number of predicted true positive values + number of predicted false positive values). For example, in the proposed algorithm, precision measures the number of the total correctly predicted positives for an asteroid family (this number also includes true negatives) in contrast to the total number of positives of that family. The equation of recall is as follows:

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

Here, *TP* represents True Positive and *TN* represents True Negative, where *TP* is the number of true positives and *FN* is the number of false negatives.

5.2.3 F1 score

F1 score is the harmonic mean of precision and recall score that is a cumulative index of prediction performance [37]. Its values can range from 0 to 1. The F1 score is calculated as follows:

$$F1\ score = \frac{2(PR)}{P + R} \quad (3)$$

Where *P* = Precision, *R* = Recall.

5.2.4 ROC-AUC

The ROC Curve (Receiver Operating Characteristic Curve) is a graph/plot of true positive rate or precision vs. false positive rate that visualizes the performance of a classification model at different classification thresholds ranging from 0 to 1 (0 indicates all wrong predictions and 1 indicates all right predictions) [38]. This performance is an indication of how accurately a model can predict a true positive value

given that the model has a certain probability/fraction of predicting a false positive value. It is also an indicator of how much higher a model ranks a random true positive case than that of a random negative case. The equations of true positive rate (Recall) and false positive rates are represented in Table 2.

AUC (Area Under Curve) is the area under the ROC curve that indicates the aggregate measure of a model's prediction performance across a wide range of classification thresholds. It indicates the cumulative probability of a model placing true positive predictions higher than false positive predictions. In general, it shows the quality of prediction across all the ranges rather than in absolute threshold points. This analysis of the area under the curve enables us to qualitatively assess the model's generalizability to new data [39]. AUC is achieved by integrating the ROC curve across the range (0, 1).

5.3 Data resampling techniques

Various imbalance data handling techniques have been investigated in this study such as Random Under Sampling (RUS), Synthetic Minority OverSampling Technique (SMOTE), using weights by cost-sensitive learning, etc.

Random UnderSampling (RUS) tackles class imbalance by reducing the number of majority class samples. This is to create a balance between majority and minority sample sizes. As the name suggests, it randomly removes samples from the majority class to closely match the number of minority class samples. This helps balance the data but sometimes leads to a loss of information from the majority class.

The Synthetic Minority OverSampling Technique (SMOTE) generates new synthetic examples for the minority class. It creates new samples by interpolating between existing minority class samples, increasing the presence of the minority class in the dataset. SMOTE generates realistic new samples, which helps the model learn better from imbalanced data without overfitting [12]. Unlike simple duplication, SMOTE resembles more real data.

Cost-sensitive learning tries to fix the problem of class imbalance by assigning higher costs for misclassifying minority class samples. In this approach resampling the dataset is not necessary. This method focuses more on influencing the training of the model. Cost-sensitive models try to reduce the overall cost thus making the model more sensitive to the minority class. This allows the model to handle imbalanced datasets by considering the costs of misclassification during the learning process, which leads to better performance on the minority class [13].

5.4 Performance analysis

The performance of this proposed model was analyzed against the evaluation metrics - Precision, Recall, F1 score, ROC curve, and AUC. To account for the evaluation, the tables in [Appendix](#) show the comparison between standalone models and the proposed model for recall (Table 4), precision (Table 4), and F1 score (Table 6). The ROC curve depicted in Fig. 5 represents the true positive rate vs. the false positive rate. It is quite evident from the ROC curve that the proposed architecture has performed very well in classifying the asteroid families. 103 families were classified

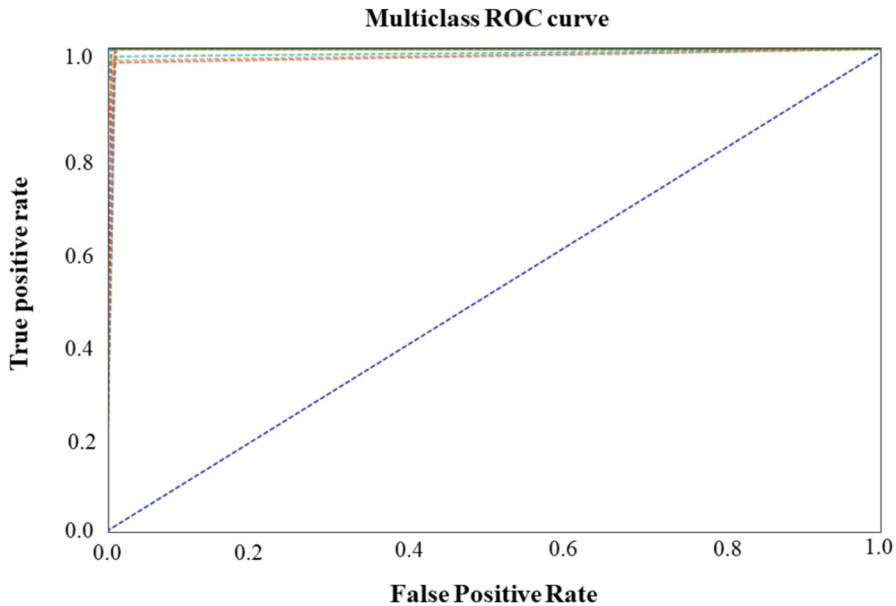


Fig. 5 ROC curve of all asteroid families

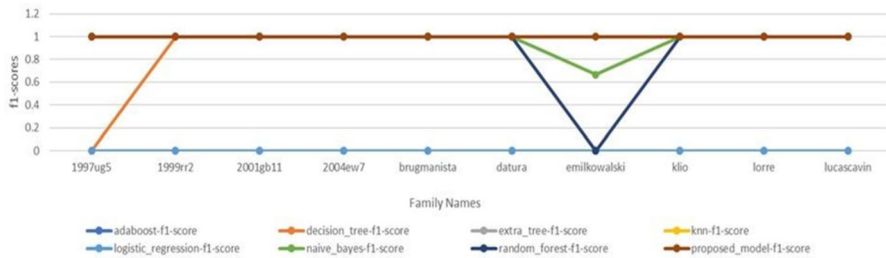


Fig. 6 F1 scores of tiny family

with a perfect AUC of 1.00 and 8 asteroid families with an AUC score of 0.99. The performance based on F1 score is shown in Fig. 6 for tiny asteroid families, Fig. 7 for small asteroid families, Fig. 8 for a lower group of medium asteroid families, Fig. 9 for a higher group of medium asteroid families, and Fig. 10 for large asteroid families. To compare, the proposed model against the works of [8] which used artificial neural networks to class (in percentage) is shown in Table 3.

The average f1 scores for all the models used in this work for the different family types has been shown in Fig. 11.

5.5 Comparison with baseline

The primary goal of developing this model was to better categorize families with limited sample numbers with more recall and precision. Both of these can be gauged

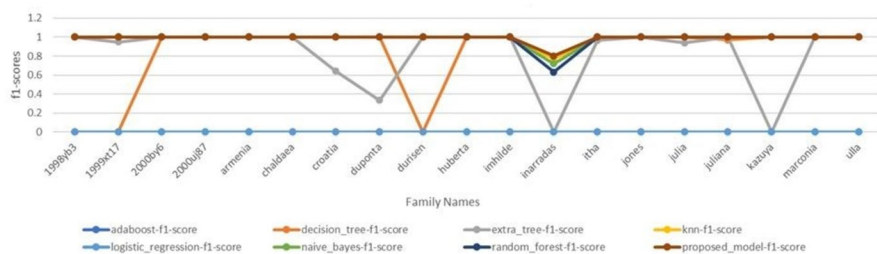


Fig. 7 F1 scores of small family

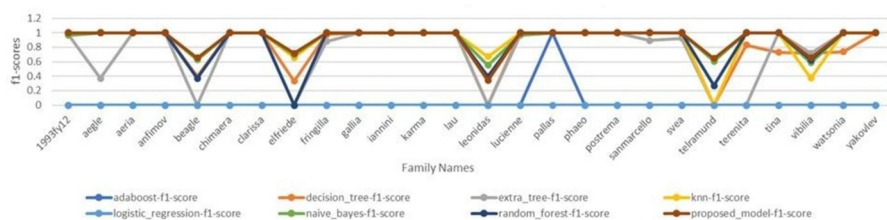


Fig. 8 F1 scores of lower group of medium family

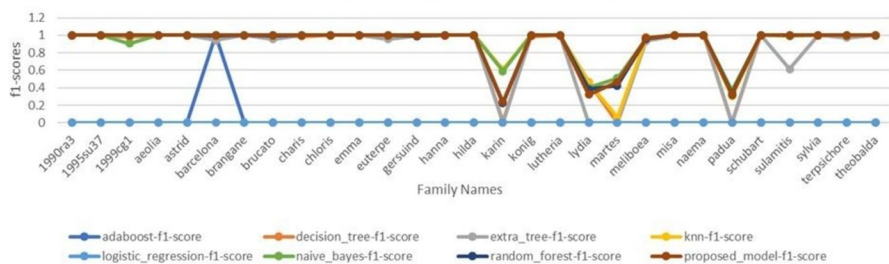


Fig. 9 F1 scores of higher group of medium family

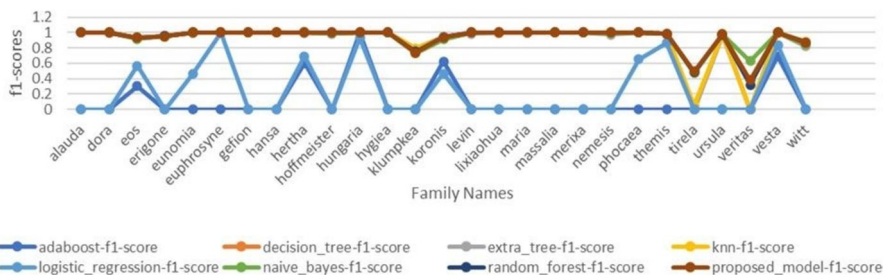


Fig. 10 F1 scores of large family

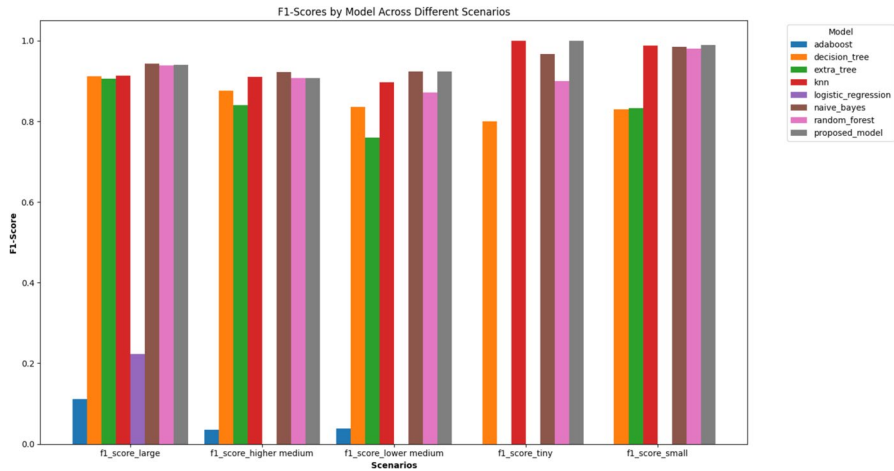


Fig. 11 Family group wise average F1 scores

using the F1 score (Table 6) metric. The proposed model achieved a perfect F1 score of 1 for all the tiny families (Fig. 6). KNN classifier also achieved an F1 score of 1 for the tiny families. All other models had a decent performance for tiny families except the AdaBoost classifier, Extra Tree classifier, and logistic regression. These three models could not correctly identify any of the tiny families and showed an F1 score of 0. Among the small families (Fig. 7), the proposed model achieved an F1 score of 1 for all families except the Inarradas. Random Forest Classifier, KNN classifier, and Naïve Bayes Classifier also performed the same as the proposed model for those families. For the Inarradas family, the proposed model got an F1 score of 0.8 which was the best among all the classifiers used. KNN classifier and Naïve Bayes classifier got F1 scores of 0.769 and 0.72 respectively for the Inarradas family. AdaBoost classifier and Logistic Regression failed to classify any of the small families correctly because of the presence of a very small amount of train data for those families. All other models showed average performance for small families. So, the proposed model clearly performed best in terms of F1 score for both the tiny families and the small families.

For the lower group of medium families (Fig. 8), the proposed model showed a perfect F1 score of 1 for 21 asteroid families. Naïve Bayes, Random Forest and KNN classifier got an F1 score of 1 for 19, 22 and 22 families respectively. In the case of the Leonidas family, the proposed model got an F1 score of 0.338 whereas the KNN classifier showed an F1 score of 0.672. Naïve Bayes and Random Forest Classifier also showed better performance in the Leonidas family than the suggested model. Except for the Leonidas family, the proposed model showed better F1 scores for the other lower group of medium families. In some of the higher group medium sized families (Fig. 9), some standalone classifiers performed better than the proposed model, especially when the sample size was larger. For the Karin family which has 895 samples, the proposed model got an F1 score of 0.2383 whereas the KNN classifier achieved 0.61 which is comparatively better. Almost all the models

except the KNN classifier and Naïve Bayes classifier performed poorly for the Karin families. KNN classifier also outperformed the proposed model in the case of the Lydia family. The proposed scored 0.322 in terms of F1 score for Lydia family. The KNN classifier and Naïve Bayes classifier scored 0.456 and 0.398 respectively. For the other higher group of medium families, the suggested model performed similarly to KNN and Naïve Bayes classifiers and better than the rest of the other models.

Among the large families (Fig. 10), the proposed model achieved a perfect F1 score of 1 for 17 families out of 29 families. For the Veritas family having 1802 members, the Naïve Bayes classifier outperformed the proposed model. The Naïve Bayes classifier showed a 0.627 F1 score whereas the proposed model got 0.382. Except for Naïve Bayes and Random Forest classifiers, all other models got an F1 score of 0 for the Veritas family. AdaBoost classifier and Logistic Regression constantly performed poorly in identifying families. In some of the lower groups of medium sized families where the number of samples is comparatively less, the proposed model outperformed all the classifiers and achieved the best result. For the Telramund family with 214 samples, the proposed model showed a 0.646 F1 score. Naïve Bayes scored 0.60 and Random Forest got 0.26. The other model got an F1 score of 0 for the Telramund family. It is evident that the proposed model outperformed the standalone machine learning models for categorizing smaller families. The proposed model performed similarly to the KNN classifier and Naïve Bayes classifier for families with large sample sizes. Overall, the proposed model had a perfect F1 score for 90 asteroid families.

5.6 Comparison with different imbalanced data handling techniques

In this section, the proposed model is compared with commonly used imbalanced data handling techniques. Here, the results of the families that have less than 250 samples have been compared only as the proposed model focuses on identifying small asteroid families. The comparison is done with other oversampling, under-sampling, cost-sensitive learning, and ensemble learning techniques such as Smote, Random Sampling (RUS), Weighted Random Forest (weighted RF), and Balanced Random Forest (BRF). The family wise F1 scores are considered as the evaluation metric.

In Fig. 12, it is observed that the proposed model achieved a perfect F1 score of 1 for all the tiny families. Smote and weighted RF also scored a perfect 1 while BRF and RUS failed to classify most of the tiny families perfectly. The scores of small families are illustrated in Fig. 13 in which the proposed model also outperformed all other techniques utilized in this work. The novel method predicted all the small families perfectly except it scored 0.8 for Inarradas family. Both Smote and weighted RF failed to classify the Duponta and Inarradas families perfectly. For the lower group of medium families shown in Fig. 14, the performance of all the methods is comparatively poorer than those of tiny and small families. Here, the proposed model showed almost similar results as weighted RF and Smote. All three techniques could not identify five of the lower groups of medium families properly. Weighted RF performed best among these three methods with an average F1 score

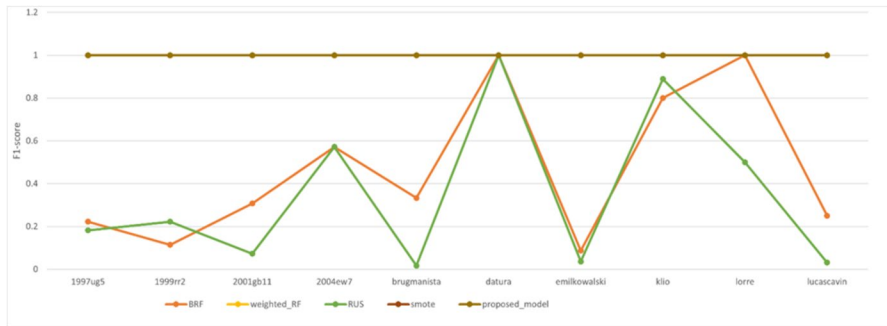


Fig. 12 Comparison among imbalance data handling techniques for tiny family

of 0.925. The proposed model scored 0.923 while smote scored 0.921. BRF and RUS constantly failed to perform well compared to the other three methods. So, comparing the results among the families having less than 250 samples, the proposed model performed comparably to the weighted RF technique and better than the other techniques. However, smote too achieved results that are not too poor compared to the proposed technique while RUS and BRF could not identify most of the families perfectly.

5.7 Comparison with previous work

For classification methods using machine learning, the works of [9] and [8] discuss machine learning algorithms and artificial neural networks respectively, which coincide with the field of work in this paper. The first study used standalone models, bagging classifiers, and boosting methods to approach the classification problem. In the study, 21 families were examined, and the classification of families with lower absolute magnitude ($H 14$) and larger inclination ($\sin(i) > 0.3$) was the main objective. This study's main flaw is that it investigates fewer families (21) than this paper does (112 families), which is more asteroid families. Moreover, the analysis of the

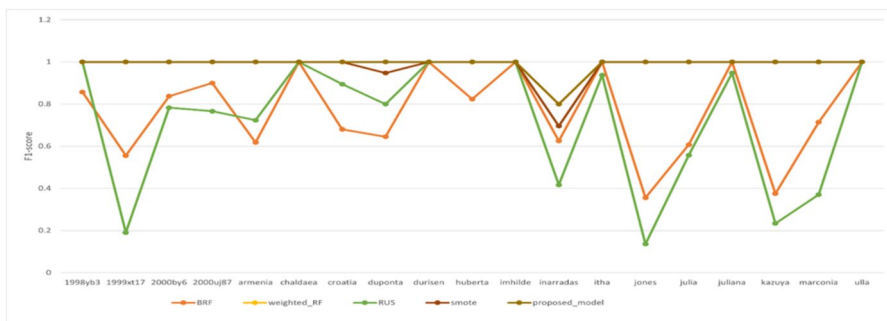


Fig. 13 Comparison among imbalance data handling techniques for a small family



Fig. 14 Comparison among imbalance data handling techniques for lower medium family

viability of the machine learning algorithms in the first article was also reproduced in this paper using the same algorithms and was compared against the proposed model.

The second article approaches the classification problem using Artificial Neural Network (ANN) models. The work also aimed at a generalized classification of the families, like that of this paper. Hence it is a good candidate to compare the results of this paper as the data source is the same and the work approaches with ANN, a classification technique that differs from the initial approach of machine learning models but generates the same estimative result. The proposed model in this paper performed better in classifying the group of families, i.e. (members < 250) than that of the second article. It must be kept in mind that the dataset was more recent compared to the work it is compared with. For further comparison, we reproduced the ANN model of [8] with the exact parameters and evaluated it on our dataset. Table 3 represents a detailed comparison of the F1 scores between the ANN model and the proposed two-step model. The results show a significant performance gap, with the proposed model achieving an average F1 score of 0.941 compared to 0.856 for the ANN model. Furthermore, the proposed model outperformed the ANN model in 38 individual families in terms of F1-score. Besides, for 12 individual families, the ANN model couldn't identify any of the test samples and achieved a 0 score. These findings highlight the superior performance and robustness of the proposed model over the ANN model developed by [8].

5.8 Limitations of the proposed model

Some larger families have suffered since their F1 score (Table 6; Fig. 10) has declined. It was discovered that, in certain instances, the F1 score of the proposed model for some families is inferior to that of other methods employed in this work. The following

families illustrate this trend: Witt, where the F1 score with K Nearest Neighbor classifier is 0.8799 and with the proposed model it is 0.8664, eos, where the F1 score with KNN classifier is 0.9356 and with this proposed model it is 0.9229, erigone, where the F1 score with Random Forest is 0.9529 and with this proposed model it is 0.9499, Klumpkea, where the F1 score with KNN classifier is 0.7921 and with this proposed model it is 0.73. In most instances, the difference in the results is not substantial. Nevertheless, it is clear that the difference is a result of the suggested model's new feature. This model's inability to categorize any new asteroid family if it does not appear in the training data is another drawback. Thus, this constraint exists regardless of the technique that uses a traditional machine learning model. Along with this the added complexity that comes with using a twostep architecture may increase the training and inference time. However, not requiring any real time implementations leaves room for the inference time of this architecture to be higher than the standard models used in this work.

6 Conclusion and future works

Classification of asteroid families has proven to be challenging over time. It has been particularly difficult to classify small families using machine learning approaches. Clearly, if the sample size used to train the algorithms is sufficiently high, machine learning approaches are quite effective for this purpose. That is not always the case, however. This is why the suggested model can be useful because it has shown that it can carry out this work efficiently. In addition, it should be highlighted that precision and recall are metrics that have been measured in the tests, but that these measurements do not necessarily reveal the truth. For this specific scenario, the F1 score (Table 6; Figs. 6, 7, 8, 9 and 10) metric is preferable since it combines both precision and recall into a single metric. This proposed model has performed well compared to earlier models for this parameter, and in some cases even better, when identifying families with fewer members. The proposed model also showed its competence in handling class imbalance, which is a significant problem in this case. It performed comparable to the weighted Random Forest technique for handling class imbalance and better than the other sampling techniques used in this study. Compared to the HCM approach, this procedure is significantly faster and more effective. Consequently, this suggested model can circumvent the constraints of traditional machine learning techniques, statistical methodologies, and HCM. However, this model like any other supervised machine learning model will fail to classify asteroid families that belong to a completely new family. It should be noted that the hyper-parameter optimization has been done with RandomSearchCV. This optimization technique does not consider all possible combinations of hyper-parameters. This is why there is a possibility to find better hyper-parameters for each algorithm. This might be an area for future research. Again, to extract additional features the AFP dataset might be cross-referenced with other datasets such as the Dark Energy Survey (DES) and taxonomical SDSS-MOC4 information. This inclusion of additional features might improve the performance of Machine Learning models to classify smaller asteroid families. Furthermore, a stable version of this classification method is being developed which would be able to handle live feed of asteroid data.

Appendix

Table 3 Comparative analysis of F1 score between proposed two step model and ANN Model

Family Name	F1-Score (ANN model)	F1-Score (proposed model)
1990ra3	1	1
1993fy12	1	1
1995su37	1	1
1997ug5	1	1
1998yb3	1	1
1999cg1	1	1
1999rr2	1	1
1999xt17	1	1
2000by6	1	1
2000uj87	1	1
2001gb11	0	1
2004ew7	1	1
adeona	0.99908341	1
aegle	0.77777778	1
aeolia	1	1
aeria	1	1
agnia	0.991011236	0.99944
alauda	1	1
anfimov	1	1
armenia	1	1
astrid	1	1
barcelona	0.967032967	1
beagle	0.43373494	0.65455
brangane	1	1
brucato	0.941176471	1
brugmanista	1	1
chaldaea	1	1
charis	0.993589744	1
chimaera	1	1
chloris	1	1
clarissa	1	1
croatia	1	1
datura	1	1
dora	1	1
duponta	0.952380952	1
durisen	1	1
elfriede	0	0.71642
emilkowalski	0	1

Table 3 (Continued)

Family Name	F1-Score (ANN model)	F1-Score (proposed model)
emma	1	1
eos	0.93569491	0.93295
erigone	0.95283773	0.94992
eunomia	0.999588647	1
euphrosyne	1	1
euterpe	0.989010989	1
fringilla	1	1
gallia	0.866666667	1
geffion	1	1
gersuind	0.994923858	1
hanna	1	1
hansa	1	1
hertha	0.999304106	1
hilda	0.737588652	1
hoffmeister	0.996858639	0.99896
huberta	0.482758621	1
hungaria	1	1
hygiea	1	1
iannini	1	1
imhilde	1	1
inarradas	0.428571429	0.8
itha	1	1
jones	1	1
julia	1	1
juliana	1	1
karin	0	0.2383
karma	1	1
kazuya	1	1
klio	1	1
klumpkea	0.786407767	0.73795
konig	1	1
koronis	0.942164782	0.9411
lau	1	1
leonidas	0	0.33898
levin	1	1
lixiaohua	1	1
lorre	1	1
lucascavin	0	1
lucienne	1	1
lutheria	0.990990991	1
lydia	0.335483871	0.32277
marconia	1	1

Table 3 (Continued)

Family Name	F1-Score (ANN model)	F1-Score (proposed model)
maria	1	1
martes	0	0.46429
massalia	0.998650472	1
meliboea	0.951351351	0.97143
merixa	1	1
misa	1	1
naema	1	1
nemesis	0.98440208	0.99827
padua	0	0.31638
pallas	0.978723404	1
phaeo	1	1
phocaea	1	1
postrema	1	1
sanmarcello	1	1
schubart	0	1
sulamitis	1	1
svea	1	1
sylvia	0.87394958	1
telramund	0	0.64662
terenita	1	1
terpsichore	1	1
themis	0.980143642	0.98376
theobalda	1	1
tina	1	1
tirela	0	0.49408
ulla	1	1
ursula	0.972436604	0.97803
veritas	0	0.38179
vesta	1	1
vibilia	0.71641791	0.65487
watsonia	1	1
witt	0.831908832	0.86649
yakovlev	1	1
Avg F1_score	0.85648339	0.941480973

Table 4 Recall scores

Family-name	AdaBoost recall	Decision tree recall	Extra tree recall	KNN recall	Logistic regression recall	Naïve Bayes recall	Random forest recall	Proposed model recall
1990RA3	0	1	1	1	0	1	1	1
1993FY12	0	1	0.96667	1	0	0.93333	1	1
1995SU37	0	1	1	1	0	1	1	1
1997UG5	0	0	0	1	0	1	1	1
1998YB3	0	0	1	1	0	1	1	1
1999CG1	0	1	0.97468	1	0	1	1	1
1999RR2	0	1	0	1	0	1	1	1
1999XT17	0	0	0.9	1	0	1	1	1
2000BY6	0	1	1	1	0	1	1	1
2000UJ87	0	1	1	1	0	1	1	1
2001GB11	0	1	0	1	0	1	1	1
2004EW7	0	1	0	1	0	1	1	1
ADEONA	0	0.99817	0.98901	1	0	0.99267	1	1
AEGLE	0	1	0.22727	1	0	1	1	1
AEOLIA	0	1	1	1	0	1	1	1
AERIA	0	1	1	1	0	1	1	1
AGNIA	0	1	0.98871	1	0	0.98194	1	1
ALAUDA	0	1	1	1	0	1	1	1
ANFIMOV	0	1	1	1	0	1	1	1
ARMENIA	0	1	1	1	0	1	1	1
ASTRID	0	1	1	1	0	1	1	1
BARCELONA	1	1	0.95	1	0	1	0.99286	1
BEAGLE	0	0.36111	0	0.97222	0	0.91667	0.27778	1

Table 4 (continued)

Family-name	AdaBoost recall	Decision tree recall	Extra tree recall	KNN recall	Logistic regression recall	Naïve Bayes recall	Random forest recall	Proposed model recall
BRANGANE	0	1	1	1	0	1	1	1
BRUCATO	0	1	0.96875	1	0	1	1	1
BRUGMANISTA	0	1	0	1	0	1	1	1
CHALDAEA	0	1	1	1	0	1	1	1
CHARIS	0	0.97452	1	1	0	1	1	1
CHIMAERA	0	1	1	1	0	1	1	1
CHLORIS	0	1	1	1	0	1	1	1
CLARISSA	0	1	1	1	0	1	1	1
CROATIA	0	1	0.47059	1	0	1	1	1
DATURA	0	1	0	1	0	1	1	1
DORA	0	1	1	1	0	1	1	1
DUPONTA	0	1	0.2	1	0	1	1	1
DURISEN	0	0	1	1	0	1	1	1
ELFRIEDE	0	0.25	0	0.875	0	0.95833	0	1
EMILKOWALSKI0	0	0	0	1	0	1	0	1
EMMA	0	1	1	1	0	1	1	1
EOS	1	1	1	1	1	0.84482	0.9676	0.94918
ERIGONE	0	1	1	0.99664	0	0.94295	0.96812	0.9547
EUNOMIA	0	1	1	1	1	1	1	1
EUPHROSYNE	0	1	1	1	1	1	1	1
EUTERPE	0	1	0.91304	1	0	1	1	1
FRINGILLA	0	0.95833	0.79167	1	0	1	1	1
GALLIA	0	1	1	1	0	1	1	1

Table 4 (continued)

Family-name	AdaBoost recall	Decision tree recall	Extra tree recall	KNN recall	Logistic regression recall	Naïve Bayes recall	Random forest recall	Proposed model recall
GEFION	0	0.99687	1	1	0	1	1	1
GERSUIND	0	1	1	1	0	0.9898	1	1
HANNA	0	1	1	1	0	1	1	1
HANSA	0	1	1	1	0	1	1	1
HERTHA	0.66794	1	0.99965	1	1	0.99965	1	1
HILDA	0	1	1	1	0	1	1	1
HOFFMEISTER	0	0.9833	0.99791	1	0	0.99374	1	0.99791
HUBERTA	0	1	1	1	0	1	1	1
HUNGARIA	1	1	1	1	1	1	1	1
HYGIEA	0	1	1	1	0	1	1	1
IANNINI	0	1	1	1	0	1	1	1
IMHILDE	0	1	1	1	0	1	1	1
INARRADAS	0	1	0	1	0	0.9	0.6	1
ITHA	0	1	0.93333	1	0	1	1	1
JONES	0	1	1	1	0	1	1	1
JULIA	0	1	0.88235	1	0	1	1	1
JULIANA	0	0.94444	1	1	0	1	1	1
KARIN	0	0.00559	0	0.87709	0	0.96648	0.15084	0.15642
KARMA	0	1	1	1	0	1	1	1
KAZUYA	0	1	0	1	0	1	1	1
KLIO	0	1	0	1	0	1	1	1
KLUMPKEA	0	1	1	0.99118	0	0.73545	0.78307	0.7425
KONIG	0	1	1	1	0	1	1	1

Table 4 (continued)

Family-name	AdaBoost recall	Decision tree recall	Extra tree recall	KNN recall	Logistic regression recall	Naïve Bayes recall	Random forest recall	Proposed model recall
KORONIS	0.99657	0.99794	1	0.87791	1	0.83882	0.97462	0.9808
LAU	0	1	1	1	0	1	1	1
LEONIDAS	0	0	0	1	0	0.60526	0.34211	0.26316
LEVIN	0	1	1	1	0	1	1	1
LIXIAOHUA	0	0.99558	1	1	0	1	1	1
LORRE	0	1	0	1	0	1	1	1
LUCASCAVIN	0	1	0	1	0	1	1	1
LUCIENNE	0	1	0.93182	1	0	0.95455	1	1
LUTHERIA	0	1	1	1	0	1	1	1
LYDIA	0	0.96894	0	0.73292	0	0.52174	0.43478	0.34783
MARCONIA	0	1	1	1	0	1	1	1
MARIA	0	1	1	1	0	1	1	1
MARTES	0	0	0	0.0339	0	0.54237	0.35593	0.44068
MASSALIA	0	1	0.99865	1	0	0.99797	1	1
MELIBOEAE	0	0.94444	0.97778	0.93333	0	0.93333	0.96667	0.94444
MERIXA	0	1	1	1	0	1	1	1
MISA	0	1	1	1	0	1	1	1
NAEMA	0	1	1	1	0	1	1	1
NEMESIS	0	1	0.9827	1	0	0.9827	1	0.99654
PADUA	0	0	0	0.2795	0	0.40373	0.34783	0.34783
PALLAS	1	1	1	1	0	1	1	1
PHAEON	0	1	1	1	0	1	1	1
PHOCAEA	0	1	1	1	0.48469	1	1	1

Table 4 (continued)

Family-name	AdaBoost recall	Decision tree recall	Extra tree recall	KNN recall	Logistic regression recall	Naïve Bayes recall	Random forest recall	Proposed model recall
POSTREMA	0	1	1	1	0	1	1	1
SANMARCELLO	0	1	0.8125	1	0	1	1	1
SCHUBART	0	1	1	1	0	1	1	1
SULAMITIS	0	1	0.44286	1	0	0.98571	1	1
SVEA	0	1	0.85714	1	0	1	1	1
SYLVIA	0	1	1	1	0	1	1	1
TELRAMUND	0	0	0	0	0	0.95349	0.2093	1
TERENITA	0	1	0	1	0	1	1	1
TERPSICHORE	0	1	0.9434	1	0	1	1	1
THEMIS	0	0.9857	1	0.96636	1	0.96972	0.99327	0.96804
THEOBALDA	0	1	1	1	0	1	1	1
TINA	0	0.57692	1	1	0	1	1	1
TIRELA	0	0	0	0.0301	0	0.46154	0.45151	0.48829
ULLA	0	1	1	1	0	1	1	1
URSULA	0	0.98643	1	0.95701	0	0.95701	0.99774	0.95701
VERITAS	0	0	0	0	0	0.96944	0.23056	0.30278
VESTA	1	1	1	1	1	1	1	1
VIBILJA	0	1	1	0.22917	0	0.54167	0.6875	0.77083
WATSONIA	0	1	1	1	0	1	1	1
WITT	0	0.76515	0.97348	0.78662	0	0.76641	0.84091	0.83586
YAKOVLEV	0	1	1	1	0	1	1	1

Table 5 Precision

Family-name	AdaBoost precision	decision tree precision	extra-tree preci- sion	KNN precision	logistic regression preci- sion	Naïve bayes precision	random forest precision	proposed model precision
1990RA3	0	1	1	1	0	1	1	1
1993FY12	0	1	1	1	0	1	1	1
1995SU37	0	1	1	1	0	1	1	1
1997UG5	0	0	0	1	0	1	1	1
1998YB3	0	0	1	1	0	1	1	1
1999CG1	0	0.95181	1	1	0	0.83158	1	1
1999RR2	0	1	0	1	0	1	1	1
1999XT17	0	0	1	1	0	1	1	1
2000BY6	0	1	1	1	0	1	1	1
2000UJ87	0	1	1	1	0	1	1	1
2001GB11	0	1	0	1	0	1	1	1
2004EW7	0	1	0	1	0	1	1	1
ADEONA	0	1	1	1	0	1	1	1
AEGLE	0	1	1	1	0	1	1	1
AEOLIA	0	1	1	1	0	1	1	1
AERIA	0	1	1	1	0	1	1	1
AGNIA	0	1	0.98206	1	0	0.97643	1	0.99887
ALAUDA	0	1	1	1	0	1	1	1
ANFIMOV	0	1	1	1	0	1	1	1
ARMENIA	0	1	1	1	0	1	1	1
ASTRID	0	1	1	1	0	1	1	1
BARCELONA	1	1	0.93662	1	0	1	1	1
BEAGLE	0	0.43333	0	0.46667	0	0.47826	0.55556	0.48649

Table 5 (continued)

Family-name	AdaBoost precision	decision tree precision	extra-tree preci- sion	KNN precision	logistic regression preci- sion	Naïve bayes precision	random forest precision	proposed model precision
BRANGANE	0	1	1	1	0	1	1	1
BRUCATO	0	1	0.94656	1	0	1	0.99225	1
BRUGMANISTA	0	1	0	1	0	1	1	1
CHALDAEA	0	1	1	1	0	1	1	1
CHARIS	0	1	1	1	0	1	1	1
CHIMAERA	0	1	1	1	0	1	1	1
CHLORIS	0	1	1	1	0	1	1	1
CLARISSA	0	1	1	1	0	1	1	1
CROATIA	0	1	1	1	0	1	1	1
DATURA	0	1	0	1	0	1	1	1
DORA	0	1	0.99112	1	0	1	1	1
DUPONTA	0	1	1	1	0	1	1	1
DURISEN	0	0	1	1	0	1	1	1
ELFRIEDE	0	0.5	0	0.525	0	0.54762	0	0.55814
EMILKOWALSKI0	0	0	0	1	0	0.5	0	1
EMMA	0	1	1	1	0	1	1	1
EOS	0.1784	0.87863	0.86848	0.87916	0.39462	0.99478	0.90121	0.91727
ERIGONE	0	0.90992	0.9044	0.91244	0	0.95416	0.93821	0.94518
EUNOMIA	0	0.99918	0.987	1	0.29889	0.99672	1	1
EUPHROSYNE	0	1	1	1	0.99481	1	1	1
EUTERPE	0	1	1	1	0	1	1	1
FRINGILLA	0	1	1	1	0	1	1	1
GALLIA	0	1	1	1	0	1	1	1

Table 5 (continued)

Family-name	AdaBoost precision	decision tree precision	extra-tree preci- sion	KNN precision	logistic regression preci- sion	Naïve bayes precision	random forest precision	proposed model precision
GEFION	0	1	0.99844	1	0	0.99687	1	1
GERSUIND	0	0.9899	0.9899	1	0	1	0.9899	1
HANNA	0	1	1	1	0	1	1	1
HANSA	0	1	1	1	0	1	1	1
HERTHA	0.54954	1	0.99584	1	0.52871	0.99896	1	1
HILDA	0	1	1	1	0	1	1	1
HOFFMEISTER	0	1	0.97154	1	0	0.9896	1	1
HUBERTA	0	1	1	1	0	1	1	1
HUNGARIA	1	1	1	1	0.8757	1	1	1
HYGIEA	0	1	0.9992	1	0	1	1	1
IANNINI	0	1	1	1	0	1	1	1
IMHILDE	0	1	1	1	0	1	1	1
INARRADAS	0	0.66667	0	0.625	0	0.6	0.66667	0.66667
ITHA	0	1	1	1	0	1	1	1
JONES	0	1	1	1	0	1	1	1
JULIA	0	1	1	1	0	1	1	1
JULIANA	0	1	1	1	0	1	1	1
KARIN	0	0.25	0	0.46866	0	0.42402	0.42188	0.5
KARMA	0	1	1	1	0	1	1	1
KAZUYA	0	1	0	1	0	1	1	1
KLIO	0	1	0	1	0	1	1	1
KLUMPKEA	0	0.65323	0.6378	0.65962	0	0.72145	0.73026	0.73345
KONIG	0	0.98131	1	1	0	1	1	1

Table 5 (continued)

Family-name	AdaBoost precision	decision tree precision	extra-tree preci- sion	KNN precision	logistic regression preci- sion	Naïve bayes precision	random forest precision	proposed model precision
KORONIS	0.45606	0.891	0.89065	0.9831	0.30218	0.99512	0.90337	0.90449
LAU	0	1	1	1	0	1	1	1
LEONIDAS	0	0	0	0.50667	0	0.51111	0.46429	0.47619
LEVIN	0	1	0.96656	1	0	1	1	1
LIXIAOHUA	0	1	1	1	0	1	1	1
LORRE	0	1	0	1	0	1	1	1
LUCASCAVIN	0	1	0	1	0	1	1	1
LUCIENNE	0	1	1	1	0	1	1	1
LUTHERIA	0	1	1	1	0	1	1	1
LYDIA	0	0.30769	0	0.33146	0	0.32184	0.34483	0.30108
MARCONIA	0	1	1	1	0	1	1	1
MARIA	0	1	0.99052	1	0	0.99761	1	1
MARTES	0	0	0	0.5	0	0.48485	0.525	0.49057
MASSALIA	0	1	0.99932	1	0	0.99932	1	1
MELIBOEAE	0	1	0.89796	1	0	0.98824	0.95604	1
MERIXA	0	1	1	1	0	1	1	1
MISA	0	1	0.9931	1	0	1	1	1
NAEMA	0	1	1	1	0	1	1	1
NEMESIS	0	1	0.96599	1	0	1	1	1
PADUA	0	0	0	0.33582	0	0.34574	0.3237	0.29016
PALLAS	1	1	1	1	0	1	1	1
PHAEOS	0	1	1	1	0	1	1	1
PHOCAEA	0	1	1	1	1	1	1	1

Table 5 (continued)

Family-name	AdaBoost precision	decision tree precision	extra-tree preci- sion	KNN precision	logistic regression preci- sion	Naïve bayes precision	random forest precision	proposed model precision
POSTREMA	0	1	1	1	0	1	1	1
SANMARCELLO	0	1	1	1	0	1	1	1
SCHUBART	0	1	1	1	0	1	1	1
SULAMITIS	0	1	1	1	0	1	1	1
SVEA	0	1	1	1	0	1	1	1
SYLVIA	0	1	1	1	0	1	1	1
TELAMUND	0	0	0	0	0	0.44565	0.375	0.47778
TERENITA	0	0.71429	0	1	0	1	1	1
TERPSICHORE	0	1	1	1	0	1	1	1
THEMIS	0	0.98075	0.97061	0.99913	0.7671	0.9974	0.97846	1
THEOBALDA	0	1	1	1	0	1	1	1
TINA	0	1	1	1	0	1	1	1
TIRELA	0	0	0	0.64286	0	0.47917	0.52326	0.5
ULLA	0	1	1	1	0	1	1	1
URSULA	0	0.96035	0.93843	0.99296	0	0.99764	0.94839	1
VERITAS	0	0	0	0	0	0.46348	0.5092	0.51659
VESTA	0.52701	1	0.98635	1	0.72319	0.99965	1	1
VIBILJA	0	0.55814	0.55814	1	0	0.63415	0.56897	0.56923
WATSONIA	0	0.5873	1	1	0	1	1	1
WITT	0	0.98537	0.70669	0.9984	0	0.93818	0.90244	0.89946
YAKOVLEV	0	1	1	1	0	1	1	1

Table 6 F1 score

Family-name	AdaBoost f1-score	decision tree-f1- score	extra-tree f1-score	KNN-f1- score	logistic regression f1-score	Naïve scorebayes-f1- score	random forest-f1- score	proposed model-f1- score
1990RA3	0	1	1	1	0	1	1	1
1993FY12	0	1	0.98305	1	0	0.96552	1	1
1995SU37	0	1	1	1	0	1	1	1
1997UG5	0	0	0	1	0	1	1	1
1998YB3	0	0	1	1	0	1	1	1
1999CG1	0	0.97531	0.98718	1	0	0.90805	1	1
1999RR2	0	1	0	1	0	1	1	1
1999XT17	0	0	0.94737	1	0	1	1	1
2000BY6	0	1	1	1	0	1	1	1
2000UJ87	0	1	1	1	0	1	1	1
2001GB11	0	1	0	1	0	1	1	1
2004EW7	0	1	0	1	0	1	1	1
ADEONA	0	0.99908	0.99448	1	0	0.99632	1	1
AEGLE	0	1	0.37037	1	0	1	1	1
AEOLIA	0	1	1	1	0	1	1	1
AERIA	0	1	1	1	0	1	1	1
AGNIA	0	1	0.98538	1	0	0.97918	1	0.99944
ALAUDA	0	1	1	1	0	1	1	1
ANFIMOV	0	1	1	1	0	1	1	1
ARMENIA	0	1	1	1	0	1	1	1
ASTRID	0	1	1	1	0	1	1	1
BARCELONA	1	1	0.94326	1	0	1	0.99642	1
BEAGLE	0	0.39394	0	0.63063	0	0.62857	0.37037	0.65455

Table 6 (continued)

Family-name	AdaBoost fl-score	decision tree-fl- score	extra-tree fl-score	KNN-fl- score	logistic regression fl-score	Naïve score	random forest-fl- score	proposed model-fl- score
BRANGANE	0	1	1	1	0	1	1	1
BRUCATO	0	1	0.95753	1	0	1	0.99611	1
BRUGMANISTA	0	1	0	1	0	1	1	1
CHALDAEA	0	1	1	1	0	1	1	1
CHARIS	0	0.9871	1	1	0	1	1	1
CHIMAERA	0	1	1	1	0	1	1	1
CHLORIS	0	1	1	1	0	1	1	1
CLARISSA	0	1	1	1	0	1	1	1
CROATIA	0	1	0.64	1	0	1	1	1
DATURA	0	1	0	1	0	1	1	1
DORA	0	1	0.99554	1	0	1	1	1
DUPONTA	0	1	0.33333	1	0	1	1	1
DURISEN	0	0	1	1	0	1	1	1
ELFRIEDE	0	0.33333	0	0.65625	0	0.69697	0	0.71642
EMILKOWALSKI0	0	0	0	1	0	0.66667	0	1
EMMA	0	1	1	1	0	1	1	1
EOS	0.30278	0.9354	0.92961	0.93569	0.56591	0.91368	0.93322	0.93295
ERIGONE	0	0.95284	0.9498	0.95269	0	0.94852	0.95293	0.94992
EUNOMIA	0	0.99959	0.99346	1	0.46023	0.99836	1	1
EUPHROSYNE	0	1	1	1	0.9974	1	1	1
EUTERPE	0	1	0.95455	1	0	1	1	1
FRINGILLA	0	0.97872	0.88372	1	0	1	1	1
GALLIA	0	1	1	1	0	1	1	1

Table 6 (continued)

Family-name	AdaBoost fl-score	decision tree-fl- score	extra-tree fl-score	KNN-fl- score	logistic regression fl-score	Naïve score	random forest-fl- score	proposed model-fl- score
GEFION	0	0.99843	0.99922	1	0	0.99844	1	1
GERSUIND	0	0.99492	0.99492	1	0	0.99487	0.99492	1
HANNA	0	1	1	1	0	1	1	1
HANSA	0	1	1	1	0	1	1	1
HERTHA	0.60299	1	0.99774	1	0.69171	0.9993	1	1
HILDA	0	1	1	1	0	1	1	1
HOFFMEISTER	0	0.99158	0.98455	1	0	0.99167	1	0.99896
HUBERTA	0	1	1	1	0	1	1	1
HUNGARIA	1	1	1	1	0.93373	1	1	1
HYGIEA	0	1	0.9996	1	0	1	1	1
IANNINI	0	1	1	1	0	1	1	1
IMHILDE	0	1	1	1	0	1	1	1
INARRADAS	0	0.8	0	0.76923	0	0.72	0.63158	0.8
ITHA	0	1	0.96552	1	0	1	1	1
JONES	0	1	1	1	0	1	1	1
JULIA	0	1	0.9375	1	0	1	1	1
JULIANA	0	0.97143	1	1	0	1	1	1
KARIN	0	0.01093	0	0.61089	0	0.58944	0.22222	0.2383
KARMA	0	1	1	1	0	1	1	1
KAZUYA	0	1	0	1	0	1	1	1
KLIO	0	1	0	1	0	1	1	1
KLUMPKEA	0	0.79024	0.77885	0.79211	0	0.72838	0.75574	0.73795
KONIG	0	0.99057	1	1	0	1	1	1

Table 6 (continued)

Family-name	AdaBoost fl-score	decision tree-fl- score	extra-tree fl-score	KNN-fl- score	logistic regression fl-score	Naïve score	random forest-fl- score	proposed model-fl- score
KORONIS	0.62575	0.94144	0.94216	0.92754	0.46411	0.91031	0.93764	0.9411
LAU	0	1	1	1	0	1	1	1
LEONIDAS	0	0	0	0.67257	0	0.55422	0.39394	0.33898
LEVIN	0	1	0.98299	1	0	1	1	1
LIXIAOHUA	0	0.99778	1	1	0	1	1	1
LORRE	0	1	0	1	0	1	1	1
LUCASCAVIN	0	1	0	1	0	1	1	1
LUCIENNE	0	1	0.96471	1	0	0.97674	1	1
LUTHERIA	0	1	1	1	0	1	1	1
LYDIA	0	0.46707	0	0.45648	0	0.3981	0.38462	0.32277
MARCONIA	0	1	1	1	0	1	1	1
MARIA	0	1	0.99524	1	0	0.99881	1	1
MARTES	0	0	0	0.06349	0	0.512	0.42424	0.46429
MASSALIA	0	1	0.99899	1	0	0.99865	1	1
MELIBOEA	0	0.97143	0.93617	0.96552	0	0.96	0.96133	0.97143
MERIXA	0	1	1	1	0	1	1	1
MISA	0	1	0.99654	1	0	1	1	1
NAEMA	0	1	1	1	0	1	1	1
NEMESIS	0	1	0.97427	1	0	0.99127	1	0.99827
PADUA	0	0	0	0.30508	0	0.37249	0.33533	0.31638
PALLAS	1	1	1	1	0	1	1	1
PHAEO	0	1	1	1	0	1	1	1
PHOCAEA	0	1	1	1	0.65292	1	1	1

Table 6 (continued)

Family-name	AdaBoost fl-score	decision tree-fl- score	extra-tree fl-score	KNN-fl- score	logistic regression fl-score	Naïve score	random forest-fl- score	proposed model-fl- score
POSTREMA	0	1	1	1	0	1	1	1
SANMARCELLO	0	1	0.89655	1	0	1	1	1
SCHUBART	0	1	1	1	0	1	1	1
SULAMITIS	0	1	0.61386	1	0	0.99281	1	1
SVEA	0	1	0.92308	1	0	1	1	1
SYLVIA	0	1	1	1	0	1	1	1
TELAMUND	0	0	0	0	0	0.60741	0.26866	0.64662
TERENITA	0	0.83333	0	1	0	1	1	1
TERPSICHORE	0	1	0.97087	1	0	1	1	1
THEMIS	0	0.98322	0.98509	0.98247	0.8682	0.98337	0.98581	0.98376
THEOBALDA	0	1	1	1	0	1	1	1
TINA	0	0.73171	1	1	0	1	1	1
TIRELA	0	0	0	0.05751	0	0.47019	0.48474	0.49408
ULLA	0	1	1	1	0	1	1	1
URSULA	0	0.97321	0.96824	0.97465	0	0.97691	0.97244	0.97803
VERITAS	0	0	0	0	0	0.62713	0.3174	0.38179
VESTA	0.69025	1	0.99313	1	0.83936	0.99982	1	1
VIBILJA	0	0.71642	0.71642	0.37288	0	0.58427	0.62264	0.65487
WATSONIA	0	0.74	1	1	0	1	1	1
WITT	0	0.86141	0.81891	0.87994	0	0.84364	0.87059	0.86649
YAKOVLEV	0	1	1	1	0	1	1	1

Author contributions Authors 1, 10, and 11 conceived the presented idea for the research. Authors 1, 3, and 4 conducted the experiments. The experimental procedures were carefully executed under their supervision while maintaining rigorous protocols. To support the research process, several personal computers (laptops) were made available by authors 1, 2, 3, 4, 6, 7, 8, and 10. 1, 2, 4, 5, 7, and 8 conducted a thorough literature review. Authors 10 and 11 supervised the project. Authors 1, 2, 3, and 4 took the lead in writing the manuscript, with support and contributions from all other authors except 10 and 11. Data analysis and graph generation were performed by authors 5, 6, 7, and 8. Authors 1, 3, and 4 contributed to the interpretation of the results achieved. All authors discussed the results and commented on the manuscript. The research was carried out with the initiative of authors 8 and 9, who are the founding members of **IUT Al-Fazari Interstellar Society (IUTFIS)**. IUT Al-Fazari Interstellar Society played a crucial role in managing the resources required for conducting the experiments. Authors 5, 6 and 7 collaborated to format the final version of the work for submission, maintaining the standards set forth by the target publication.

Data availability <http://asteroids.matf.bg.ac.rs/fam/properelements.php>.

Declarations

Competing interests The authors declare no competing interests.

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